

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

Tellurium solution 1000 ppm in ca. 20% hydrochloric acid

Tellurium solution 1000 ppm in ca. 20% hydrochloric acid

J/8327/08

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

Tel: +03-5525 7888

CHEMTREC Malaysia 1-800-815-308 (Malay)

CHEMTREC Malaysia (Kuala Lumpur) +(60)-327884561 (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Substances/mixtures corrosive to metal             | Category 1 (H290) |
| Skin Corrosion/Irritation                          | Category 2 (H315) |
| Serious Eye Damage/Eye Irritation                  | Category 2 (H319) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |

**Label Elements**


Signal Word

Warning

**Hazard Statements**

H290 - May be corrosive to metals

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H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P234 - Keep only in original packaging  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P390 - Absorb spillage to prevent material damage

### Storage

P402 - Store in a dry place  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P406 - Store in corrosion resistant polypropylene container with a resistant inliner

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors  
Toxic to terrestrial vertebrates

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Hydrochloric acid | 7647-01-0 | 20       |
| Tellurium dioxide | 7446-07-3 | 0.125    |
| Water             | 7732-18-5 | 79.875   |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

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**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors.

**Hazardous Combustion Products**

Thermal decomposition can lead to release of irritating gases and vapors, Hydrogen chloride gas.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental precautions**

Should not be released into the environment.

**Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

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## Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component         | Malaysia | ACGIH TLV                  | OSHA PEL                                                                                                             |
|-------------------|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid |          | Ceiling: 2 ppm             | Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 ppm<br>(Vacated) Ceiling: 7 mg/m <sup>3</sup> |
| Tellurium dioxide |          | TWA: 0.1 mg/m <sup>3</sup> | (Vacated) TWA: 0.1 mg/m <sup>3</sup>                                                                                 |

| Component         | European Union                                                                                               | The United Kingdom                                                                                         | Germany                                                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hydrochloric acid | TWA: 5 ppm 8 hr<br>TWA: 8 mg/m <sup>3</sup> 8 hr<br>STEL: 10 ppm 15 min<br>STEL: 15 mg/m <sup>3</sup> 15 min | STEL: 5 ppm 15 min<br>STEL: 8 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 2 mg/m <sup>3</sup> 8 hr | TWA: 2 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 3 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 2 ppm (8 Stunden). MAK<br>TWA: 3.0 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 4 ppm<br>Höhepunkt: 6 mg/m <sup>3</sup> |
| Tellurium dioxide |                                                                                                              | STEL: 0.3 mg/m <sup>3</sup> 15 min<br>TWA: 0.1 mg/m <sup>3</sup> 8 hr                                      |                                                                                                                                                                                                                                                      |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Particulates filter conforming to EN 143

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

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## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Appearance                              | Colorless                |                                   |
| Physical State                          | Liquid                   |                                   |
| Odor                                    | No information available |                                   |
| Odor Threshold                          | No data available        |                                   |
| pH                                      | 1                        |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flash Point                             | Not applicable           | Method - No information available |
| Evaporation Rate                        | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | No data available        |                                   |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Specific Gravity / Density              | 1.03                     |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Water Solubility                        | Miscible                 |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

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## Possibility of Hazardous Reactions

### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

## Conditions to Avoid

Incompatible products. Excess heat.

## Incompatible Materials

Bases. Metals.

## Hazardous Decomposition Products

Thermal decomposition can lead to release of irritating gases and vapors. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

No acute toxicity information is available for this product

#### **(a) acute toxicity;**

##### **Oral**

Based on available data, the classification criteria are not met

##### **Dermal**

Based on available data, the classification criteria are not met

##### **Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component         | LD50 Oral               | LD50 Dermal             | LC50 Inhalation       |
|-------------------|-------------------------|-------------------------|-----------------------|
| Hydrochloric acid | 238 - 277 mg/kg ( Rat ) | > 5010 mg/kg ( Rabbit ) | 1.68 mg/L ( Rat ) 1 h |
| Tellurium dioxide | LD50 > 5 g/kg ( Rat )   | -                       | -                     |
| Water             | -                       | -                       | -                     |

#### **(b) skin corrosion/irritation;**

Category 2

#### **(c) serious eye damage/irritation;**

Category 2

#### **(d) respiratory or skin sensitization;**

##### **Respiratory**

No data available

##### **Skin**

No data available

#### **(e) germ cell mutagenicity;**

No data available

#### **(f) carcinogenicity;**

No data available

There are no known carcinogenic chemicals in this product

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(g) reproductive toxicity; No data available

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system.

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component         | Freshwater Fish                                                            | Water Flea                 | Freshwater Algae | Microtox |
|-------------------|----------------------------------------------------------------------------|----------------------------|------------------|----------|
| Hydrochloric acid | 282 mg/L LC50 96 h<br>Gambusia affinis<br>mg/L LC50 48 h<br>Leuciscus idus | 56mg/L EC50 72h<br>Daphnia | -                | -        |
| Tellurium dioxide | LC50: > 100 mg/L, 96h<br>semi-static<br>(Oncorhynchus mykiss)              |                            |                  |          |

### Persistence and degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

### Bioaccumulative potential

Bioaccumulation is unlikely

### Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

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**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Solutions with low pH-value must be neutralized before discharge

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN3264  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s. Hydrochloric acid solution

### Road and Rail Transport

UN-No UN3264  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s. Hydrochloric acid solution

### IATA

UN-No UN3264  
Hazard Class 8  
Packing Group III  
Proper Shipping Name Corrosive liquid, acidic, inorganic, n.o.s. Hydrochloric acid solution

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

| Component         | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Hydrochloric acid | 231-595-7 | X    | X   | X     | X    | X    | X     | X    | KE-20189 |
| Tellurium dioxide | 231-193-1 | X    | X   | X     | X    | X    | X     | X    | KE-33097 |
| Water             | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |

| Component         | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Hydrochloric acid | 25 tonne                                                                                  | 250 tonne                                                                                |                            | Annex I - Y34                      |
| Tellurium dioxide |                                                                                           |                                                                                          |                            | Annex I - Y28                      |

### National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance



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## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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**Revision Summary**

SDS sections updated, 2, 3, 9, 15.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### **Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**